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Multilayered or Multiseriate Epidermis

Generally, epidermis is single layered, but in certain leaves, multilayered upper epidermis is present, Example: ***Ficus, Nerium, and Peperomea***.

In *Ficus* upper epidermal layer contains cystoliths made up of calcium carbonate crystals.

In *Nerium*, in the multilayered epidermis the outer layer alone is **cutinized.**

Epidermal Outgrowths

There are many types of epidermal outgrowths in stems. The unicellular or multicellular appendages that originate from the epidermal cells are called **trichomes**. Trichomes may be branched or unbranched and are one or more one celled thick. They assume many shapes and sizes. They may also be glandular (Example: *Rose, Ocimum*) or non-glandular.

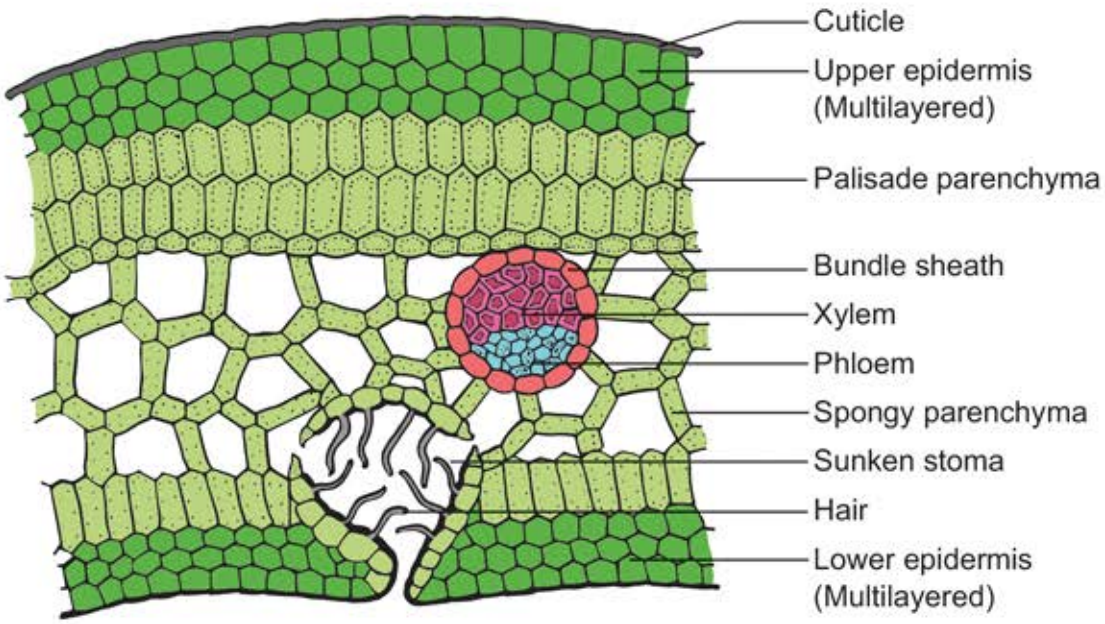


Figure 9.15: T.S. of *Nerium* Leaf



The trichomes on the leaves of insectivorous plants secrete muco-polysaccharides that trap insects.

Piliferous layer of the root has two types of epidermal cells, long cells and short cells. The short cells are called **trichoblasts**. Trichoblasts are elongate into root hairs. Epidermal hairs can also be in the form of stellate hairs (star shaped) present in plants. Example: styrax, many members of Malvaceae and Solanaceae.

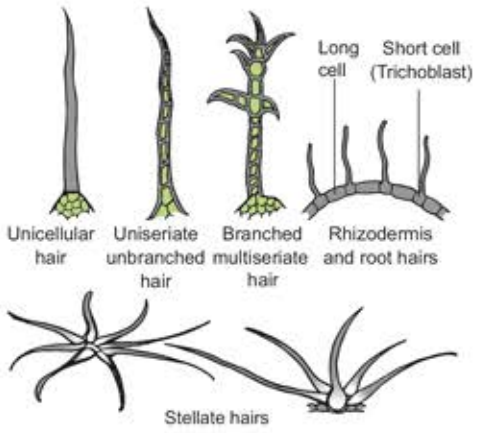


Figure 9.16: Types of Trichomes